

ANALYSTLAB AFRICA

Experience Lab Internship Programme

FinTrust Digital Bank - Week 3 Project Documentation

Data Science Track: FinTrust Predictive Risk Model - Advanced Development & Validation

Project: FinTrust Financial Intelligence & Digital Banking Support Solution

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A. Application of the Week 1-2 Foundation

What I originally planned to accomplish

In Week 1, I defined the predictive problem: whether transaction and customer information can help predict the Risk_Review_Flag. Week 2 turned that plan into a prepared modelling dataset, exploratory analysis, eight engineered features and a baseline Logistic Regression.

What Week 2 delivered

- A joined, quality-checked modelling dataset (FinTrust_Modelling_Dataset_Prepared.csv) and nine EDA visualisations.
- A baseline Logistic Regression (class_weight="balanced", 12 features, random stratified 80/20 split, threshold 0.50): ROC-AUC 0.668, Recall 0.594, Precision 0.289, F1 0.389, Accuracy 0.635.
- An honest list of open items: only one algorithm tried, no tuning, Transaction_Status and Account_Status not quantified, customer-category features kept despite little signal, and no check across customer groups.

How Week 3 builds on it

Week 3 does not start a new project. It starts from the Week 2 prepared dataset, reproduces the Week 2 baseline exactly from the submitted notebook (the code asserts that all five metrics match), re-fits the same recipe on a chronological split so every comparison is like-for-like, and then works through each Week 2 open item.

Changes to my approach and why

- **Chronological split.** The evaluation split moved from random to chronological (first 60% of transactions to train, next 20% to validation, last 20% to test) because a risk-review model would score future transactions. I checked that the random split was not optimistic (ROC-AUC 0.668 random vs 0.676 chronological).
- **Metrics.** Emphasis moved from accuracy to PR-AUC, precision and recall at stated operating points, and lift at a review budget, because Week 2 accuracy (63.5%) is below the 80.8% of an "always No" model.
- **Test set discipline.** The test set was kept for final reporting only; feature and threshold decisions used training and validation data.

B. Use of the Approved Project Resources

Resource	How it was used	What it provided	Limitation encountered
FinTrust_Modelling_Dataset_Prepared.csv (Week 2)	Starting dataset for all Week 3 work; Week 3 features were added on top of it	12,000 transactions (1 Jan - 31 Mar 2026) with customer details joined; the 96 missing Device_Type and 96 missing Location values already filled with "Unknown"	Customer attributes carry almost no signal for the target; only three months of data
Week 2 notebook and documentation	Source of the baseline recipe, features, decisions and open issues	Baseline metrics to reproduce and beat; list of unresolved issues to follow up	None
FinTrust_Data_Dictionary	Checked which customer fields are intended for modelling	Field definitions and modelling guidance	Covers customer fields only
Project Brief, Roadmap, Week 1-3 assignments	Scope, track requirements and the reminder that the target is synthetic	Business framing and the Week 3 deliverable list	Not used directly in code
FinTrust Financial Knowledge Base	Not used (GenAI track resource)	Not applicable	Not applicable

No other intern's output was available or used, and no cross-track integration is claimed.

C. Tools Used

- **Python and Jupyter Notebook** - all data preparation, analysis and modelling (Python 3.14, pandas 3.0.2, NumPy 2.4.4, scikit-learn 1.8.0).
- **Pandas / NumPy** - loading, feature engineering and calculations.
- **Scikit-learn** - pipelines, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, grid search, time-series cross-validation, permutation importance, partial dependence and evaluation metrics.
- **SciPy** - chi-square tests for the feature signal table.
- **Matplotlib / Seaborn** - visualisations.

- **Git / GitHub** - version control and repository.

D. Practical Outputs Produced

- **Updated Modelling Dataset** - fintrust_modelling_dataset_v2.csv (engineered features and train/validation/test labels). This is the only data file produced.
- **Week 3 notebook** - FinTrust_Week3_DS_Advanced_Modelling.ipynb (Parts A-G, validation evidence and summary).
- **Additional models** - Decision Tree, Random Forest and Gradient Boosting, plus extended-feature Random Forest and Gradient Boosting variants.
- **Feature refinement** - feature signal table, ablation table and feature decision table, all in the notebook.
- **Model comparison** - comparison tables at two operating points (notebook Section 5).
- **Error analysis and model interpretation** - confusion matrices, segment analysis, coefficients, permutation importance, partial dependence and tree rules, with nine saved charts.
- **Candidate final model** - candidate_model.joblib and candidate_model_card.json.
- **Assumptions, limitations and decisions log** - notebook Section 10.
- **README and requirements.txt**, and this Week 3 Project Documentation.

E. Week 3 Development Work

Part A - Review of the Week 2 baseline

Evaluated on the chronological test set (2,400 transactions, 19.2% flagged), the Week 2 recipe gives ROC-AUC 0.676, precision 0.296 and recall 0.647 at threshold 0.50, but it flags about 42% of all transactions to get there.

Issue / gap	Evidence	Required improvement
Accuracy misleading	63.5% accuracy vs 80.8% for "always No"	Report PR-AUC, precision/recall at stated thresholds and lift
Moderate ranking, weak precision	ROC-AUC about 0.67; precision about 0.29	Try better features and model families; choose the threshold deliberately
Noisy customer attributes kept	A demographics-only model has validation ROC-AUC 0.49 (chance); Account_Type_Savings ranked among the top five "risk" terms	Remove non-informative attributes
Hour entered as a straight line	Hour coefficient is negative, yet the flag rate is 28.7% in hours 00-05 vs 16.0% otherwise	Engineer a night-time flag
Raw, skewed amount as a linear term	Flag rate 17.4% below the 90th percentile (about NGN 142k) and 34.6% above it: a step	Log transform plus a high-amount flag

Issue / gap	Evidence	Required improvement
Customer counts use future rows	Customer_Transaction_Count and Transaction_Frequency are counted over the whole 3-month window (look-ahead); neither is significant ($p = 0.91$ and 0.28)	Test against past-only history features; drop if no gain
Single untuned model, random split, no uncertainty estimates	Listed in Week 2's own open items	Time-series CV tuning, more models, bootstrap confidence intervals
Status fields never quantified	Week 2 unresolved issue	Sensitivity tests

Part B - Additional models

Four classifiers were tuned with 5-fold time-series cross-validation on the training period only, scoring PR-AUC (suited to a target that is about 20% positive): Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. Two extended-feature variants (adding channel, device and hour) were also run for Random Forest and Gradient Boosting. Decision thresholds were chosen on the validation period only. The models use the refined features from Part C. Selected settings: Logistic Regression $C = 0.01$; Decision Tree depth 5 with at least 50 samples per leaf; Random Forest depth 4 with at least 10 samples per leaf; Gradient Boosting depth 2, learning rate 0.03, 200 iterations.

Part C - Feature refinement

Feature decision	Reason	Expected effect
Remove Age, Tenure, Digital Engagement, Customer Segment and Account Type	Not significant against the flag; no validation gain when added back; also removes a fairness concern	Less noise, simpler and fairer model
Remove Channel (and Device, Location) from the candidate	Not significant at $p < 0.01$; no consistent gain	Parsimony and portability
Replace linear Transaction_Hour with is_night (00:00-05:59)	The real pattern is a night-time block, not a line	Captures the time effect correctly
Transform Amount_NGN to log_amount and add high_amount (train 90th percentile, NGN 142,058)	Skewed amount; the flag rate steps up above the threshold	Stronger, more stable signal
Keep Is_International and Transaction_Type	About twice the flag rate for international; Transfer and Cash Withdrawal highest	Strong, stable signals
Remove the Week 2 whole-window customer counts	Look-ahead; no validation gain	Removes look-ahead leakage
Test and reject past-only customer history features	Leakage-safe but no consistent gain	Keeps the model simple

Feature decision	Reason	Expected effect
Keep Transaction_Status and Account_Status out (Week 2 decision retained)	May not be known at decision time	Deployment-safe

An ablation across nine feature sets shows the refined 5-feature core performs about the same as the full 12-feature Week 2 set on validation (ROC-AUC 0.659 vs 0.662 for Logistic Regression and 0.665 vs 0.659 for Gradient Boosting). Adding hour, channel/device, history, whole-window counts or demographics changes validation ROC-AUC by at most about 0.01. Only Transaction_Status adds a visible gain (+0.012 for Logistic Regression and +0.010 for Gradient Boosting on validation), and it is excluded for deployment reasons.

Part D - Model comparison (chronological test set)

At a 20% review budget (flag the top-scored 20% of transactions):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Week 2 baseline (12 features)	0.740	0.326	0.331	0.329	0.676
Logistic Regression	0.752	0.360	0.366	0.363	0.672
Decision Tree	0.741	0.337	0.357	0.347	0.674
Random Forest	0.754	0.366	0.377	0.371	0.676
Gradient Boosting	0.752	0.358	0.366	0.362	0.679
Random Forest (extended features)	0.745	0.352	0.385	0.368	0.684
Gradient Boosting (extended features)	0.749	0.355	0.372	0.364	0.691

At the F1-optimal threshold (high-recall screening mode):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Always "No"	0.808	0.000	0.000	0.000	0.500
Week 2 baseline at threshold 0.50	0.636	0.296	0.647	0.406	0.676
Logistic Regression	0.511	0.259	0.829	0.395	0.672
Decision Tree	0.518	0.262	0.829	0.398	0.674
Random Forest	0.517	0.262	0.829	0.398	0.676
Gradient Boosting	0.517	0.262	0.829	0.398	0.679
Random Forest (extended features)	0.517	0.261	0.827	0.397	0.684
Gradient Boosting (extended features)	0.510	0.260	0.838	0.397	0.691

Trade-offs

- No single metric decides. Accuracy is misleading on an imbalanced target.
- On ranking (ROC-AUC) every model and the Week 2 baseline are within about 0.02 of each other. Paired bootstrap confidence intervals for the differences all include zero, except that the extended Gradient Boosting leans positive (+0.015, 95% CI -0.003 to +0.033).
- At a fixed 20% review budget the new models beat the baseline on precision and recall. For Logistic Regression, Random Forest and Gradient Boosting the intervals exclude zero.
- The Decision Tree is the most readable but the weakest (PR-AUC 0.299, below the baseline).
- The F1-optimal mode flags 52-62% of all transactions and does not beat the Week 2 baseline on F1 (the candidate scores 0.395 vs 0.406), so the 20%-budget operating point is the recommended default.

Part E - Error analysis (Logistic Regression candidate, 20% review budget)

Confusion matrix on the test set: 1,637 true negatives, 301 false positives, 293 false negatives and 169 true positives.

- **The model effectively counts risk signals** (night-time, international, high amount, Transfer or Cash Withdrawal). Transactions with two or more signals are always flagged; with none, never; with one, about 3% of the time.
- **False negatives are "quiet" transactions.** All 293 have one signal or none, with a median amount of about NGN 9k (true positives: about NGN 57k). Yet the actual flag rate is 8.6% with zero signals and 21% with one signal. These labels are not explainable from the available fields, so they look like irreducible noise in the synthetic target.
- **False positives look almost identical to true positives** (night-time 59% vs 60%, international 19% vs 18%). With two signals the true flag rate is only about 35%, so roughly two-thirds of flagged transactions are false alarms, which sets a precision ceiling of about 0.36.
- **By segment:** false-positive rates are highest for international (84%), amounts above NGN 300k (73%, only 42% of them truly flagged) and 100-300k (46%), and night-time (41%). Recall is only 9-10% for Bill Payment, Airtime/Data, Card Purchase and Deposit because those types are low-risk.

Conclusion: the remaining errors are a data limitation rather than a model limitation, so more model complexity is unlikely to fix them.

Part F - Model interpretation

- **Logistic Regression coefficients (the candidate model's own):** Transfer (odds ratio about 1.59), night-time (1.39), Cash Withdrawal (1.30), high amount (1.29) and international (1.24) raise predicted risk; Card Purchase, Bill Payment, Deposit and Airtime/Data lower it. Strong regularization shrinks the coefficients, so direction and ranking matter more than size.
- **Permutation importance** (drop in validation PR-AUC): amount is largest (log_amount for tree models, high amount for the linear model; one underlying factor), followed by transaction type (0.019-0.039), night-time (0.019-0.031) and international (0.005-0.026).
- **Decision-tree rules** split first on Transfer, then on night-time and an amount boundary of roughly NGN 160k, consistent with the EDA.

- **Consistency with Week 2:** the directions match the Week 2 coefficients. Week 2 ranked Is_International first by coefficient size, but its importance is lower because international transactions are only about 4% of the data.
- These are associations with a synthetic label, not causes, and not evidence about real fraud.

Part G - Candidate final model (carried into Week 4)

Item	Detail
Model selected	Logistic Regression (C = 0.01, balanced class weights)
Reason	A selection rule fixed on validation and cross-validation data before reading the test table: among the four core-feature models keep those within 0.01 of the best validation ROC-AUC and PR-AUC, then take the simplest. Logistic Regression, Random Forest and Gradient Boosting were eligible, so the simplest, Logistic Regression, was chosen. Differences between models are within noise, so this is a choice of simplicity and explainability, not a claim of superiority. The rule is borderline for Logistic Regression: its validation ROC-AUC (0.658) is almost exactly 0.01 below the best (0.668).
Features used	Transaction_Type, log_amount, high_amount (at or above NGN 142,058), is_international, is_night
Preprocessing	One-hot encoding of Transaction_Type (unknown-safe); standard scaling of the numeric features; balanced class weights; thresholds learned on training and validation data only
Evaluation (test, 20% budget)	Accuracy 0.752, precision 0.360, recall 0.366, F1 0.363, ROC-AUC 0.672, PR-AUC 0.321, lift 1.85x
Evaluation (test, F1-optimal mode)	Recall 0.829, precision 0.259, F1 0.395, accuracy 0.511
Limitations	Ranking no better than Week 2; modest ceiling; scores not calibrated; synthetic target; short window; test set viewed once; candidate choice sensitive to the 0.01 tolerance

F. Testing / Evaluation of the Work

Work completed	Test evaluation	Finding	Action / improvement	Result
Baseline reproduction	Re-ran the exact Week 2 code and asserted the metrics	Reproduced 0.635 / 0.289 / 0.594 / 0.389 / 0.668	Used as the reference for all comparisons	Like-for-like baseline
Improvement vs baseline	Paired bootstrap (1,000 resamples) on the test set	ROC-AUC change - 0.004 (95% CI -0.022 to +0.015); precision at 20% budget +0.034 (CI +0.008 to +0.060); recall +0.035 (CI +0.004 to +0.069)	Reported both results honestly	No evidence of better ranking; moderate evidence of better use of a limited review budget

Work completed	Test evaluation /	Finding	Action improvement /	Result
Leakage check	Shuffled-label test (10 shuffles)	Mean test ROC-AUC 0.498 (range 0.42 to 0.58)	None needed	No leakage or artefact
Stability over time	Rolling-origin cross-validation (5 folds)	ROC-AUC 0.670 +/- 0.014	None needed	Stable
Customer memorisation	Customer-grouped vs random 5-fold CV	0.671 vs 0.672	None needed	No reliance on recognising customers
Threshold stability	Validation-chosen 20% threshold applied to test	Flags 20.0% of validation vs 19.6% of test; precision 0.367 vs 0.360	None needed	Threshold transfers well
Split integrity and past-only features	ID overlap, time ordering, manual recomputation	No overlap; ordering holds; recomputation matches	None needed	Passed
Model reload	Saved with joblib, reloaded and re-scored	Identical test predictions	None needed	Reproducible
Transaction_Status sensitivity	Added to the candidate	Test ROC-AUC +0.002	Kept out (not known at decision time)	Negligible cost of the Week 2 decision
Account_Status sensitivity	Added to the candidate	Test ROC-AUC +0.001 (flag rate Active 18.9%, Dormant 20.7%, Restricted 23.9%)	Kept out; revisit only if availability is confirmed	Week 2 open issue resolved
Customer-group consistency	Model flag rate across gender, segment and account type	0.18 to 0.23 across groups; small samples	Documented	Not conclusive either way

Required improvement chain (Initial output -> Test -> Finding -> Refinement -> Re-test -> Result). Week 2 baseline (balanced Logistic Regression, 12 features) -> evaluated on the chronological test set -> ranking moderate, 42% of transactions flagged, only 33% recall and 33% precision at a 20% budget, with noise attributes, a wrongly shaped hour term and look-ahead counts -> five engineered features, tuned models and validation-chosen operating points -> re-ran on the same test set with bootstrap intervals -> the same ranking from 5 inputs instead of 12, with no look-ahead, and at the same 20% budget precision 0.326 to 0.360 and recall 0.331 to 0.366.

G. Important Decisions

Decision	Reason	Evidence	Effect on the work
Chronological 60/20/20 split with time-series CV	The model would score future transactions	Random vs chronological ROC-AUC 0.668 vs 0.676	Realistic, leak-free evaluation
Drop customer attributes	No signal and a fairness concern	Significance tests, ablation, demographics-only AUC 0.49	Simpler, fairer model
Replace linear hour with a night flag	The pattern is a block	28.7% vs 16.0% flag rate	Correct time effect
Remove whole-window customer counts	Look-ahead; unavailable for new customers	No validation gain	Removes leakage
Keep Transaction_Status and Account_Status out	May not be known at decision time	+0.002 and +0.001 test ROC-AUC if added	Deployment-safe
Select the candidate by a pre-stated rule	Avoid choosing on the test set	Validation and CV metrics; rule picked Logistic Regression	Defensible choice (borderline rule noted)
Report two operating points; default to the 20% budget	The F1-optimal mode flags over half of all transactions	F1 0.395 vs baseline 0.406	Practical default
No cross-track integration claimed	No other intern's output was used	Not applicable	Honest reporting

H. Assumptions, Limitations, Risks & Unresolved Issues

What remains relevant from Week 2

- The target is synthetic and must not be described as real fraud.
- Customer attributes add little (now confirmed three ways: significance tests, ablation and a demographics-only model).
- Transaction_Status and Account_Status stay excluded.

What has been resolved

- Whether other algorithms help: tested, with no meaningful ranking gain.
- Whether the random split inflated results: it did not.
- The effect of Transaction_Status and Account_Status: quantified.
- Checks across customer groups: done, with small samples.

What has changed

- Evaluation protocol (chronological), feature set (12 to 5, look-ahead removed) and metric emphasis (PR-AUC, precision and recall at operating points, lift).

New issues

- There is no ranking improvement over the Week 2 baseline; gains appear only at a fixed review budget and in simplicity.
- The F1-optimal threshold does not beat the Week 2 baseline on F1 (0.395 vs 0.406).
- Error floor: false negatives carry no signal in the available fields.

New risks

- The test set has been viewed once for all comparisons, so Week 4 claims should rely on rolling-origin CV and bootstrap intervals.
- The extended-feature Gradient Boosting scored highest (test ROC-AUC 0.691) but the gain is not conclusive.
- The candidate rule is borderline for Logistic Regression (0.01 tolerance), so a different library version or machine could select another model; the Week 4 write-up should state the choice explicitly.

Technical and methodological limitations

- Scores are not calibrated probabilities; the high-amount threshold is data-dependent.
- Three-month window; 2,400-row test set (ROC-AUC interval about ± 0.027); no customer-level hold-out; many comparisons were made.

Effect on Week 4

Use the same evaluation protocol for comparability, report confidence intervals, avoid quoting accuracy, and describe the work as a simpler, safer model that uses a limited review budget better - not as a more accurate fraud detector.

Week 3 Project Summary

1. **What I planned to accomplish.** Take the Week 2 baseline (balanced Logistic Regression, 12 features) and develop it into a stronger, validated candidate: review it, refine features, train and compare additional models, analyse errors, interpret the model and choose a candidate for Week 4.
2. **What I completed.** All Parts A-G of the Data Science track, plus an exact reproduction of the Week 2 baseline, an updated modelling dataset, a saved candidate model and model card, and this documentation.
3. **Major development activities.** Chronological split; time-series CV tuning; engineered `log_amount`, `high_amount` and `is_night`; removed eight Week 2 inputs (customer attributes, look-ahead counts, channel) and replaced the linear hour term with a night flag; four tuned models plus two extended variants; two operating points.
4. **Key findings and results.** Candidate Logistic Regression: test ROC-AUC 0.672 (Week 2 recipe 0.676, no real difference), PR-AUC 0.321. At a 20% review budget precision 0.360 and recall 0.366 vs 0.326 and 0.331 for Week 2. Main drivers: amount (a step above about NGN 142k), transaction type (Transfer, Cash Withdrawal), night-time and international.
5. **Testing and validation performed.** Baseline reproduction check, chronological hold-out, time-series CV, paired bootstrap, shuffled-label test, rolling-origin CV, customer-grouped CV, threshold stability, split-integrity and past-only-feature checks, model reload test, `Transaction_Status` and `Account_Status` sensitivity, and customer-group consistency.

6. **Improvements made.** The same ranking quality from 5 inputs instead of 12, with look-ahead and customer-attribute features removed; better precision and recall at an equal 20% review budget (+0.034 and +0.035; bootstrap intervals exclude zero); a deliberate operating-point choice instead of the default 0.50 cut-off.
7. **Major challenges.** The achievable ceiling is low: every model and feature set lands within about 0.02 ROC-AUC of the baseline; the F1-optimal threshold flags more than half of all transactions; errors concentrate where no feature carries information.
8. **Important decisions.** Chronological evaluation; drop customer attributes and look-ahead features; keep status fields out; select the candidate by a pre-stated rule; report two operating points; make no unsupported integration claims.
9. **Remaining limitations.** Modest ceiling; uncalibrated scores; short window; test set already viewed once; synthetic target.
10. **What must be completed in Week 4.** Final evaluation using rolling-origin CV and bootstrap intervals; optionally confirm whether the extended-feature Gradient Boosting is genuinely better; optional calibration; confirm the operating point against a stakeholder review budget; finalise the model card and presentation; integrate only with outputs that actually exist.
11. **Final-week priorities.** (1) Final validation of the candidate; (2) a clear, honest presentation: same ranking, simpler and safer model, better use of a limited review budget; (3) hand-off of the model, model card and modelling dataset if other tracks' outputs are available; (4) tidy the README and repository.